DC POWER DISTRIBUTION: Need To Think Of Current Situation

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In India, electrical distribution infrastructure is not well extended in rural areas. However, with advanced development in power electronics switching and control devices, it is now possible to replace an existing AC power distribution network with a DC power supply for better energy efficiency. For this, a case study of actual load data has been done at academic buildings of Matoshri College of Engineering, Nashik. Energy-efficiency analysis on different loads has shown that, if an existing AC power distribution network is replaced by a DC distribution network, considerable amount of energy can be saved.

Current scenario

The electrical power distribution system at the end of 19th century was based on DC technology. This Edison DC distribution system had the disadvantage that power-generating plants feeding heavy distribution conductors and consumer loads tapped of these. This system operated at the same voltage level throughout.

But to have a better option, a three-wire system was also used to save the cost of the copper conductor. The three wires were at 110, zero and -110 volts relative potential. 100-volt lamps could be operated between either +100 or -100-volt legs of the system and zero-volt neutral conductor that carried only unbalanced current between the positive and negative sources.

For fast-growing power electronics development, it is possible to completely replace the existing AC distribution network with DC distribution network, from the point of view of today's power system scenario at distribution side. Most of the loads on distribution side are of DC type and, unfortunately, these all are fed by an AC-type supply distribution system involving a number of energy-conversion stages.

In case of AC type there should be source synchronisation, which is not required in case of DC. DC can be drawn directly from solar, wind and the grid by the loads, depending on which source is available. Harmonic issues and phase-balancing problems are not present in a DC system. It offers a lower cost of ownership in building wiring, copper and connectors along with an increase in efficiency of eight to ten per cent, which is truly significant.

A proper distribution system offers higher efficiency and the potential to extract power from multiple available sources. Then there are benefits that are not as immediately apparent. Most back-up energy sources such as batteries and flywheels are inherently DC.

Further, telecom and server loads run on DC, so there are fewer intermediate efficiency-robbing stages, along with greater reliability due to fewer potential points of failure with the DC approach. According to Central Electricity Authority (CEA), India has a shortage of around 11 per cent

Fig. 1: A unipolar LVDC distribution system